

BOM:

#	Description	Price	V/I
2	ESP32-S3 N8R16	10	5/0.2
66	Hall Effect Sensors (A3144, unipolar digital)	5	5/0.35
4	CD74H4067 Multiplexers	5	5/~0
2	TXS0108E Logic Level Shifters	3	5/0.01
64	WS2812B SMD LEDs	11	5/3.2
2	Rotary Encoder w/ switch	2	5/~0
4	keyboard switches		5/~0
1	3,5" LCD TFT SPI Touch Screen	15	5/0.16
1	12V PSU	8	:)
x	1000 UF 6.3V capacitors	4	6.3/~0 (leakage)
32+ 18	3x10mm N52 Magnets	10	:)
8	3x5mm N52 Magnets	5	:)
2	17HS4023 stepper motors	20	12/1.4
2	TMC2226 stepper motor driver	14	6 or 12/0.015
4	P25/20 electromagnet	20	12/1.3 (only 1 at a time)
2	1N4007 schottky diode	4	
2	MS3625 microphone	5	~0
4	conical springs (dont know which yet)	5	:)
1	PTFE sheet	20	:)
1	5/7.5 A automotive fuse (for + psu terminal)		
1	Raspberry Pi 4B 4GB Ram	50	5/1.3
1	Raspberry Pi 4 Heat Sink	7	0
1	MGN12H 500mm Linear Rail	20	:)
1	MGN9H 450mm Linear Rail	22	:)
1	gt2 belt 6mm	15	:)
2	20T 6mm gt2 pulleys (5mm bore)	5	:)
2	20T W6 B5 idler pulley w/ T	4	:)
4	20T W6 B5 idler pulley w/o T	4	:)
1	12 to 5v buck converter (15/20A)	15	12V/4.8 = 5V/10
1	KW12-3 Limit Switch?	5	
4	16t gt2 6mm idlers no t (unsure about 16t or 20t)	4	
20	M3 threaded inserts (buy directly from RAMPA!)	25	
4	Arctic S4028	25	12/0.4
0.6	Multiplex board (incl. cutting service by	70	

m ²	carpenter)		
20 m	30 AWG wire (red, black, 5m blue)	7	
20 m	24 AWG wire (red, black, 5m blue)	15	
5m	18AWG wire (red, black)	7	
2m	cable sleeve (could also 3d print a drag chain)	5	
10	WAGO 221-415	10	
4	1/4W 10K resistor	4	
2	100UF 25V Capacitor (motor drivers)	2	
2	1/4W 1k resistor	2	
4	IRLZ44N MOSFET	2	
1	250x500x0.5mm ptfe sheet	10	
1	Epoxy glue	25	
1	Casting epoxy	25	
	total		5V 8.5 A 12V 7.9A